

Jackson Bopp

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Education

University of South Florida

B.S. Computer Engineering, Dean's List, Gold's Scholars | GPA: 3.52

Tampa, FL

Fall 2026

Florida SouthWestern State College

High School Diploma, Associate in Arts, Dean's List | GPA: 3.86

Punta Gorda, FL

Fall 2022

Open Source

Open Source Test and Reliability Fixes

2026

C++, Python, C, GoogleTest/gmock, pytest, Docker, Gerrit and GitHub code review

- OpenBMC phosphor-user-manager: root-caused a reported 60-of-69 test failure to a missing test-environment dependency rather than a code defect, then found and fixed two real bugs uncovered once the environment was corrected, an unhandled exception that could abort setup for every user over one bad record, and a C++ construction-order bug that let unit tests silently run against the real system instead of their mocks; verified with full before-and-after test counts across both root and non-root runs.
- Zephyr RTOS: found a devicetree macro bug in the MIPI DBI display driver, proved it with a targeted compile-time check against an existing in-tree test fixture, fixed it, and added a permanent regression test so it cannot silently reappear.
- PlatformIO: added a regression test reproducing a device-monitor crash before fixing it, then confirmed the project's full lint suite (black, isort, codespell, pylint) passed clean; merged upstream.

Projects

Testbench

Summer 2026

Python, MQTT, Docker, GitHub Actions, MSP430FR2355, ESP32, STM32F103C8, RP2040

- Built a hardware-agnostic test automation platform: a capture-and-replay tool and hardware simulator let the full test suite run without physical devices attached.
- Wrote reference firmware for four different microcontrollers emitting a shared JSON-over-UART frame format, plus a schema-validated parser and a cross-platform test suite asserting every board's output against that schema.
- Set up a GitHub Actions workflow that spins up Postgres and Mosquitto containers and runs an end-to-end simulated-hardware regression test on every push.

Radiology Second-Opinion Agent

Summer 2026

FastAPI, Docker Compose, MLflow, Evidently, pytest

- Data and MLOps engineer on a 4-person hackathon team; owned the model serving layer and its test coverage, including unit tests, mock-LLM tests, and endpoint validation, plus Evidently-based data drift monitoring in production.

Wireless Motor PWM Controller

Summer 2026

MSP430FR2355, Bare-Metal C, HM-10 BLE, L298N, I2C, UART, PWM

- Collaborated with a 4-member team to build and validate bare-metal C firmware driving a DC motor from wireless and physical keypad input, including timing verification across DCO frequency drift.
- Documented the system with a full wiring diagram, system block diagram, and demo video as part of the final project report.

Experience

City of Winter Haven Technology Services | Smart City Student Volunteer | Winter Haven, FL

April 2026 - Present

- Installed server hardware and performed rack maintenance for city technology infrastructure.
- Provisioned and configured YubiKey security keys for staff authentication.
- Resolved IT support tickets and assisted with early-stage AI hosting infrastructure.

Subs N' Such | Manager, Shift Leader, and Front-End Web Support | Lutz, FL

April 2023 - March 2026

- Built and maintained a Square-based website for online ordering.
- Managed shift operations and enforced food safety procedures.

Skills

Testing & QA: GoogleTest/gmock, pytest, regression testing, root cause analysis, flaky test diagnosis, hardware-in-the-loop test automation, capture-and-replay tooling, schema validation

CI/CD & Tooling: GitHub Actions, Docker, Gerrit code review, clang-format, checkpatch, meson, west, git bisect

Languages: Python, C, C++, TypeScript, Bash, MATLAB, Verilog/VHDL

Systems: Linux, D-Bus, MQTT, PostgreSQL, Docker Compose, Raspberry Pi, MSP430, ESP32, STM32

Certifications: Certified SOLIDWORKS Associate (CSWA), CSWA Additive Manufacturing

Involvement

Member, Society of Hispanic Professional Engineers (SHPE), Society of Asian Scientists and Engineers (SASE), and IEEE